

Principles of Economics

Class 13: Technological Progress, Institutions, Unemployment and Living Standards in the Long Run

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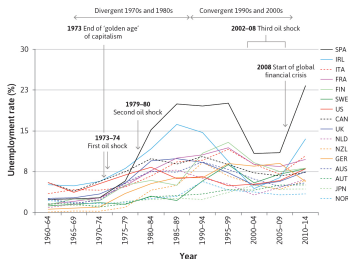
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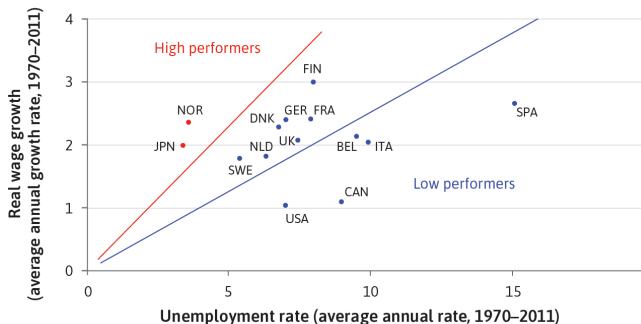
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- ▶ Today, unemployment **in the long run** (against the short- and medium-run fluctuations already seen)
- ▶ Production has become more capital-intensive since the industrial revolution...
- ▶ But unemployment has not risen everywhere (a sad exception: Spain)
- ▶ Rates were low and similar until the 1960s; the divergences appear in the 1970s.
- ▶ We will see that the differences reflect:
 - ▶ Differences in institutions and policies;
 - ▶ **Not** technological progress.



- For that, a long-run labour market model that explains why growth does not mean more unemployment in the long run...



...although it could cause it in the short run.

Job creation and unemployment [Macro U1]

Long-run labour market model [Macro U1]

The role of institutions and policies [Macro U9]

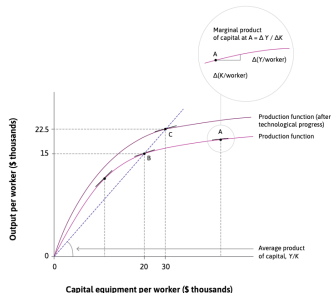
1. Job creation and unemployment [Macro U1]

- ▶ **Technological improvements allow** firms to **earn innovation rents**:
 - ▶ But firms that lag behind fail – **creative destruction** [class 1]
- ▶ **Technological progress and the accumulation of capital goods are complementary**:
 - ▶ New technologies require new machines;
 - ▶ Without innovation, investing in new machines pays less.
- ▶ **That complementarity, plus innovation rents, raises living standards.**
- ▶ Two key concepts:
 - ▶ **Capital intensity of production**: $\frac{K}{L}$, capital goods per worker.
 - ▶ **Average product of labour (APL)**: $\frac{Y}{L}$, output per worker.
- ▶ Per-worker variables focus attention on the average capital needed for a given output.

Job creation and unemployment

Technological progress and capital intensity

- ▶ Technological progress raises the Average Product of Labour (APL): output per worker.
- ▶ We represent it with a **production function**:
 - ▶ **Diminishing marginal returns to capital**: each extra unit of capital raises APL less.
- ▶ Technological progress:
 - ▶ **Shifts APL up and offsets the diminishing returns to capital.**
 - ▶ By raising the return to capital, production becomes more capital-intensive.
- ▶ The marginal product of capital (the slope) is the same at B and at C.

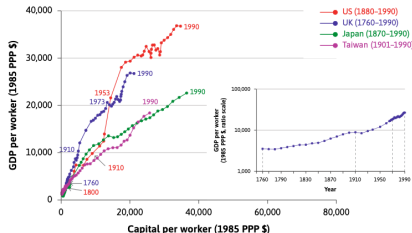


Note that this production function is **different from the one we saw in Class 1** (in which Capital was constant)

Job creation and unemployment

Technological progress and capital intensity

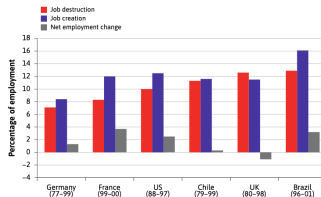
- ▶ Today's rich countries **raised output per worker as production became more capital-intensive.**
- ▶ In many countries, **capital productivity** remained **almost constant** over time.
- ▶ This looks inconsistent with the usual **concave production function.**
- ▶ But they combined **capital accumulation and technological progress:**
- ▶ And so their **production function shifted upward progressively.**



Job creation and unemployment

Job destruction and creation

- ▶ More capital per worker **could have hit employment hard**:
 - ▶ If the capital share rises, less labour is needed!
- ▶ But **labour-saving technologies can create other jobs**:
 - ▶ For example, reallocation of labour from low-tech to high-tech firms.
 - ▶ In many countries **job destruction was offset by strong job creation**:
 - ▶ The net effect is generally positive.
 - ▶ Net change in employment = job creation - job destruction



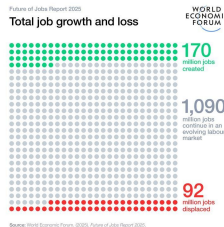
Job creation and unemployment

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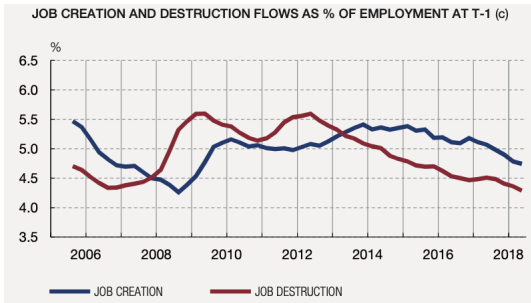
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- ▶ **Net change in employment = job creation - job destruction**



Introduction

- ▶ Job creation and destruction rates **vary with the business cycle**:
- ▶ * **Job creation is strongly pro-cyclical**:
It rises in booms and falls in recessions;
- ▶ * **While job destruction is mostly counter-cyclical**;

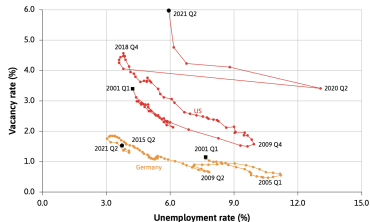


SPANISH CASE

Definition

- ▶ The relation between available jobs and unemployment, as a % of the labour force, is **the Beveridge curve**.
- ▶ **The Beveridge curve:** inverse relation between the unemployment rate and the vacancy rate.
- ▶ **Recessions:** high unemployment, low vacancy rate
- ▶ **Booms:** low unemployment, high vacancy rate
- ▶ William Beveridge (1879-1963), English economist and politician, studied unemployment and job reallocation; a founding father of the UK's Social Security system.

Beveridge curves for the US and Germany (2001 Q1–2021 Q2).



- ▶ Why are job vacancies not filled immediately if there is unemployment?
 - ▶ The labour market has matching frictions.
- ▶ **There is mismatch when available jobs do not fit unemployed workers:**
 - ▶ Workers **do not have the necessary skills**;
 - ▶ **Regional disparities**;
 - ▶ Job seekers and/or employers may **not know** each other.
- ▶ **Sharp changes in one industry, or transformations that stop workers switching, raise mismatch [lower efficiency].**
- ▶ In general, **policy and technology can improve efficiency:**
 - ▶ Over time, policy or technology can **shift the Beveridge curve** towards the origin.

Labour flows and the Beveridge curve

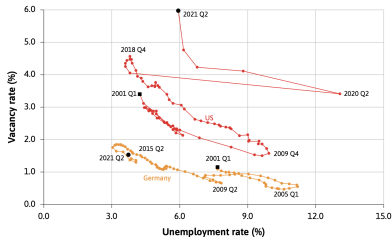
The Beveridge curve across countries and over time

- ▶ Beveridge curves give information about **labour-market differences over time and across countries**

- ▶ **Germany:** The Beveridge curve shifted towards 0 due to:

- ▶ Labour-market reforms: counselling for the unemployed and earlier cuts in benefits.

Beveridge curves for the US and Germany (2001 Q1–2021 Q2).



- ▶ **USA:** The Beveridge curve moved away from 0 due to:
 - ▶ Skills mismatch – the 2008–09 crisis especially hit the housing sector.
 - ▶ Limited mobility: the housing bubble stopped workers selling their homes and moving.

Labour flows and the Beveridge curve

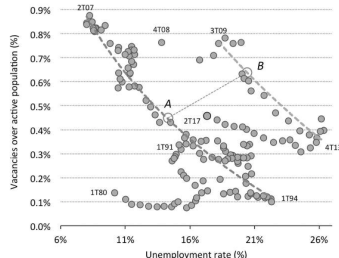
The Beveridge curve across countries and over time

- ▶ Beveridge curves give information about **labour-market differences over time and across countries**

- ▶ **The Spanish case:**

- ▶ In the 2010 Crisis it moves away from 0:
Large mismatch (construction sector)
- ▶ Gradual improvement
2014-16:
Labour reform?

Figure 7: BEVERIDGE CURVE IN SPAIN, 1980-2016



Source: Bosca, Jose E., et al. "Shifts in the Beveridge Curve in Spain and their macroeconomic effects." *Revista de Economía Aplicada* 25.75 (2017): 5-27.

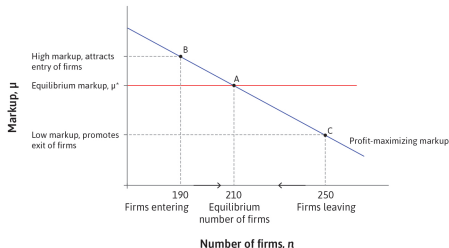
2. Long-run labour market model [Macro U1]

- ▶ For long-run unemployment we **extend the labour market model**;
- ▶ **Key difference** from the short run:
 - ▶ **Firms can enter/exit** the market:
 - ▶ Depending on market conditions, a firm may stop producing and leave.
 - ▶ If the market is profitable, firms enter.
 - ▶ And therefore, **the capital stock can change**.
- ▶ **Long-run unemployment depends** on how **policies and institutions address**:
 - ▶ Work incentives – depend on the **wage-setting curve**;
 - ▶ Investment incentives – depend on the **price-setting curve**.
- ▶ In the **long run, equilibrium**: wages, employment and the number of firms are constant.

Long-run labour market model: Elements

Number of firms in the market

- ▶ The profit ratio (=markup) determines the number of firms in the market:
 - ▶ If the markup is high: firms enter.
 - ▶ If the markup is low: firms exit.



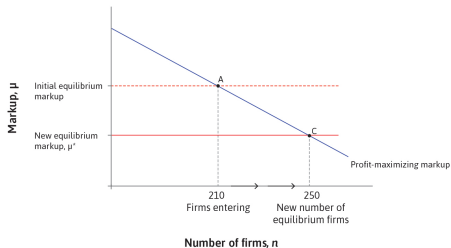
- ▶ This is a **self-correcting process**:

High markup $\Rightarrow \uparrow$ Firms $\Rightarrow \uparrow$ Competition $\Rightarrow$
 $\uparrow$ Demand elasticity $\Rightarrow \downarrow$ Markup

Long-run labour market model: Elements

Number of firms in the market

- ▶ The equilibrium markup (at which firms neither enter nor exit) can change:
- ▶ Suppose **legislation that improves the business environment** (lower expropriation risk, a commitment not to raise taxes);
- ▶ Entry becomes more profitable, and that **lowers the equilibrium markup**.



Long-run labour market model: Elements

The real wage and the price-setting curve

- ▶ Real wage (w) depends on:

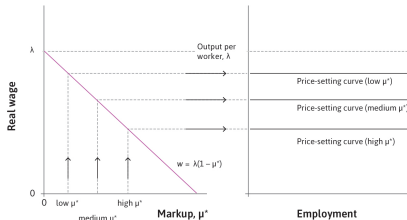
- ▶ Productivity (λ) and
- ▶ Equilibrium profit level/markup (μ^*)

- ▶ The long-run price-setting curve:

$$w = \lambda(1 - \mu^*)$$

- ▶ The price-setting curve is higher if:

- ▶ Output per worker is high
- ▶ μ^* is low:
 - ▶ $\uparrow$ competition: quality of human capital/infrastructure
 - ▶ $\downarrow$ long-run tax expectations; expropriation risk; opportunity cost of capital (e.g. low interest rates); expected material costs.



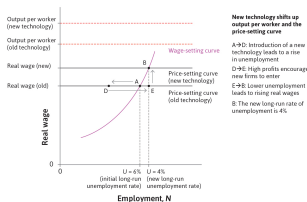
Note that both graphs are representations of the long-run price-setting curve. The different shape results from using different axes!

Long-run labour market model: Elements

Technological improvements: real wages and unemployment in the long run

- ▶ **New technologies** can raise real wages and employment in the long run:
- ▶ Suppose an economy in a **long-run equilibrium**...
... with a 6% unemployment rate
- ▶ A **new technology raises output per worker**...
...and hence the price-setting curve:
 - ▶ **A** → **D**: new technology raises unemployment;
 - ▶ **D** → **E**: high profits induce firm entry and raise employment;
 - ▶ **E** → **B**: lower unemployment raises real wages;
...long-run unemployment falls to 4%.
 - ▶ **Adjustment** takes time and **destroys jobs in the short run**.

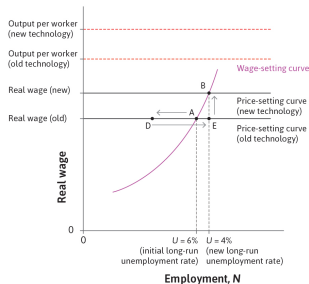
- ▶ **Adjustment gap**: lag between the external change and reaching the new equilibrium.
- ▶ **Diffusion gap**: time it takes the economy to adopt the innovation.



Long-run labour market model: Equilibrium

Technological improvements: real wages and unemployment in the long run

- ▶ In reality, **unemployment may not fall with technological progress:**
 - ▶ **The wage-setting curve may shift upward.**
- ▶ **This is because technological change may lead to:**
 - ▶ **Bargaining for a fair share by unions:**
 - ▶ Higher profits increase pressure on wages.
 - ▶ **Policies to help those who lose their jobs to technology.**
 - ▶ Greater disutility of effort;
 - ▶ Increases in the **reservation wage.**
- ▶ The higher the wage-setting curve, the likelier we end with a **higher real wage and the same unemployment [B']**.



New technology shifts up output per worker and the price-setting curve

A→D: Introduction of a new technology leads to a rise in unemployment

D→E: High profits encourage new firms to enter

E→B: Lower unemployment leads to rising real wages

B: The new long-run rate of unemployment is 4%

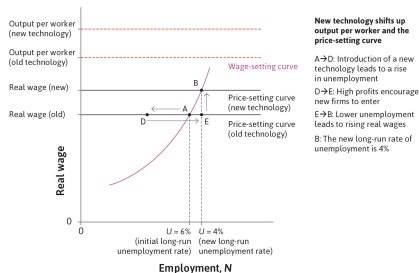
Long-run labour market model: Equilibrium

Technological improvements: short run versus long run

- ▶ We must separate the **short- and long-run effects of technological change**:

	SR	LR
Figure	A → D	A → B
Employment	↓	↑
Unemployment	↑	↓
Wage share	↓	no change
Inequality	↑	slight fall

- ▶ Number of firms and capital stock are fixed. → Net job destruction
- ▶ Output adjusts to the new equilibrium (B) [no change in the wage-setting curve]. → Net job creation



- ▶ As studied in CORE, technological change:
 - ▶ Increased inequality in the SR
 - ▶ But reduced it in the LR

What is the long run in REAL life?

What is the long run?

- ▶ **There is no exact timeline that defines "the long run":**

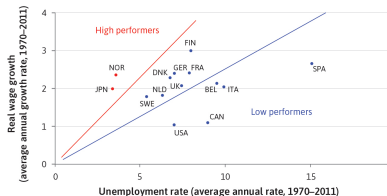
The long run is a misleading guide to current affairs. In the long run we are all dead. Economists set themselves too easy, too useless a task if in tempestuous seasons they can only tell us that when the storm is long past the ocean is flat again.
(A Tract on Monetary Reform, John Maynard Keynes, 1923)

- ▶ We separate short and long run by **whether certain circumstances hold**:
 - ▶ If some factors cannot change (capital), we are in the **short run**:
 - ▶ The long run is the **period in which everything is variable**: the economy adjusts to shocks and reaches a stable equilibrium.
- ▶ CORE details **the adjustment of the US labour market to the China import shock** (1990s):
 - ▶ Early estimates fell short: 2.4 million jobs lost between 1999 and 2011, and the market is still adjusting.

3. The role of institutions and policies [Macro U9]

Introduction

- ▶ **Policies and institutions shape** the long-run equilibrium and how long it takes to get there.
- ▶ Given the **technology shock we have studied**, in the long run the economy should:
 - ▶ The price-setting curve should shift up more than the wage-setting one; otherwise employment growth fades and inequality rises
 - ▶ **Adjustment** should be "**fast**" and **complete**, so that the whole economy benefits from technological progress.
- ▶ Between 1970 and 2011 **some countries did better than others**: the difference is **their institutions and policies**.



Differences across countries

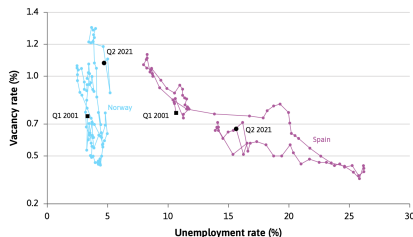
- ▶ For this shock, these **characteristics** matter:
 - ▶ **Institutions**: inclusive unions (many firms and sectors) do not use their full bargaining power because wage rises slow long-run job creation;
 - ▶ **Policies**: well-designed unemployment insurance and placement services keep unemployment low.
- ▶ **CORE details** successful cases (Norway and Japan) and unsuccessful ones (Spain):
 - ▶ Norway: inclusive unions and employers set wages by labour productivity; legislation also pushes the wage-setting curve down and expands long-run employment.
 - ▶ Japan: employer associations **coordinate wage-setting across firms**; corporations do not compete to hire, so as not to push wages up.
 - ▶ Spain: **non-inclusive unions and legislation** that protects jobs more than workers; hence a labour market that works "poorly".
- ▶ **There is no magic formula**: institutions and policies differ across countries and over time (context, cultural factors, etc.)

The role of institutions and policies

Differences across countries: Norway versus Spain

- ▶ **Inclusive unions** in Norway moderated "unsustainable" wage increases (avoiding high unemployment);
- ▶ **Active labour market policies** cut mismatch and unemployment (its extent and duration); in Spain they seem to have been insufficient.
- ▶ Their **Beveridge curves** show the difference.

Beveridge curves for Spain and Norway
(2001 Q1–2021 Q2)



Norway's Beveridge curve is closer to the origin than Spain's: less unemployment for the same level of vacancies.

Changes over time

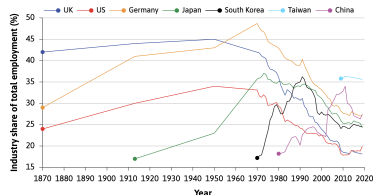
- ▶ **Institutions and policies** make the difference in employment and wage growth; and they are **not static**, least of all in the long run;
- ▶ **Changes in (less efficient) policies can help** reach a long-run equilibrium faster;
 - ▶ However, **they are often difficult** because they create winners and losers. **Social consensus is not always easy.**
- ▶ **The Netherlands and the UK** (neither success nor failure) saw unemployment rise in the 1970s from higher fuel prices and greater worker bargaining power.
- ▶ Both **countries managed to shift the wage-setting curve down**:
 - ▶ The Netherlands did so by making **institutions more inclusive** (as in Norway);
 - ▶ The UK did so by **reducing the power of (non-inclusive) unions**.
- ▶ Margaret Thatcher, British Prime Minister (1979-1990), was key in cutting union power in the UK. More on this impact, [here](#).



The role of institutions and policies

Changes over time: Economic structure

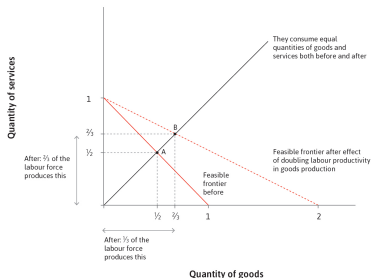
- ▶ Adjustment to shocks also **depends on the structure of economic activity**.
- ▶ Today, **fewer than 1 in 20 workers in rich economies works in agriculture**.
- ▶ As **countries become richer**:
 - ▶ The share of labour devoted to **(agricultural/industrial) production decreases**;
 - ▶ Many workers **move to the services sector**.
- ▶ This has **consequences for future productivity growth**.
- ▶ By **structure of the economy** we mean how many workers are in each sector.



The role of institutions and policies

Changes over time: Economic structure

- ▶ How does technological change move the **labour force across the major sectors?**
- ▶ Suppose an economy with a **unit labour supply (=1)** to produce goods and services.
 - ▶ Feasible set = maximum quantity that can be produced.
- ▶ At **point A** the labour force is split equally between goods and services (50%)



- ▶ Higher industrial productivity **expands the feasible set:**
 - ▶ **You can now produce twice as many goods** with the same inputs [with no change in services productivity]
 - ▶ If consumption patterns do not change, the economy moves to B.

- ▶ **The labour force shifts** from goods production to services production: Before 1/2 goods 1/2 services, now 1/3 goods, 2/3 services

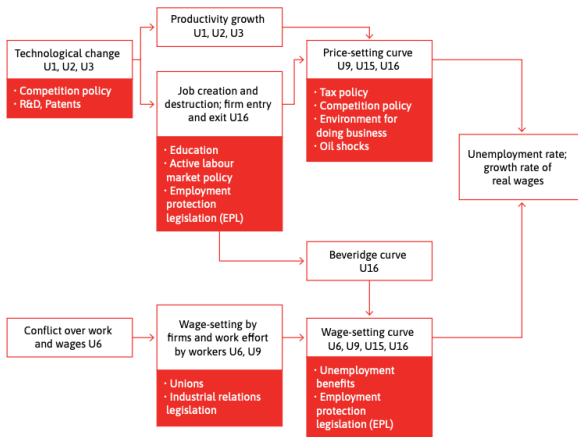
Important factors not included in the previous model:

- ▶ **Productivity increase in services:**
 - ▶ Digitalisation raises productivity in services such as music distribution and digital information.
 - ▶ Recent advances in AI suggest improvements above all in services.
- ▶ **The substitution of goods for services in consumption:**
 - ▶ if the relative price of goods falls, their demand and hence their production will increase.
 - ▶ For example: buying washing machines versus going to the laundromat to wash clothes.
- ▶ **Rising demand for services as income increases.**
 - ▶ For example, buying ingredients to bake a cake or buying it ready-made.
- ▶ **Import and export patterns.**
 - ▶ For example, the US produces technology research / China manufactures products [Apple and iPhones]

The role of institutions and policies

Changes over time: Economic structure

- Determinants, institutions, policies and shocks affecting long-run unemployment:



What you take home from each section:

1. **Introduction** → unemployment rates were similar until the 1960s and diverged afterwards: the cause is institutions and policies, not technology.
2. **Job creation and unemployment** → technological progress and capital accumulation are complementary: they raise output per worker, destroy jobs and create others; the net effect is generally positive.
3. **Labour flows and the Beveridge curve** → vacancies and unemployment move inversely; the position of the curve measures matching efficiency, and it shifts (Germany towards the origin, the US and Spain away from it).
4. **Long-run labour market model** → firms enter and exit, so the markup and the capital stock adjust: the long-run price-setting curve is $w = \lambda(1 - \mu^*)$ and technology can lower unemployment, with an adjustment gap and a diffusion gap on the way.
5. **The role of institutions and policies** → inclusive unions and active labour market policies decide how much of the technological gain reaches employment and wages: Norway and Japan against Spain; there is no magic formula.

Key concepts: capital intensity, average product of labour, creative destruction, job creation and destruction, Beveridge curve, matching efficiency, mismatch, markup, entry and exit of firms, long-run price-setting curve, adjustment gap, diffusion gap, inclusive unions, active labour market policies, structural change.